

TØ 6 – Mikroøkonomi 2 – 2024

Problem 1

Consider the following static game of incomplete information. Nature selects the type θ of player 1, where $\theta = 2$ with probability $2/3$ and $\theta = 0$ with probability $1/3$. Player 1 observes θ (he knows his own type), but player 2 does not observe θ . Then the players make simultaneous and independent choices and receive payoffs as described by the following matrix.

1\2	X	Y
A	0,1	1,0
B	1,0	$\theta,1$

- Draw the Bayesian normal form matrix of this game.
- Compute the Bayesian Nash equilibrium.

This implies that the game can either be with $P(\theta = 2) = \frac{2}{3}$

	X	Y
A	0,1	1,0
B	1,0	2,1

Or with $P(\theta = 0) = \frac{1}{3}$

	X	Y
A	0,1	1,0
B	1,0	0,1

Consider strategy (AB, X) :

With a probability of $P(\theta = 2) = \frac{2}{3}$, player 1 is the first type and $P(\theta = 0) = \frac{1}{3}$ he is type 2.

In other words, if player 1 is the first type, she chooses A, giving him a payoff of 0, and if she is the second type, she chooses B, giving her a payoff of 1

We calculate expected payoff in by:

$$0 \cdot \frac{2}{3} + 1 \cdot \frac{1}{3} = \frac{1}{3}$$

And likewise, player 2's expected payoff:

$$1 \cdot \frac{2}{3} + 0 \cdot \frac{1}{3} = \frac{2}{3}$$

And so on

	X	Y
AA	0,1	1,0
AB	$\frac{1}{3}, \frac{2}{3}$	$\frac{2}{3}, \frac{1}{3}$
BA	$\frac{2}{3}, \frac{1}{3}$	$\frac{5}{3}, \frac{2}{3}$
BB	1,0	$\frac{4}{3}, 1$

Compute the Bayesian Nash equilibrium.

This is done by IESDS:

λ^2	X	Y	
AA	0, 1	1, 0	2.
AB	$\frac{1}{3}, 1$	$\frac{2}{3}, \frac{1}{3}$	1.
BA	$\frac{2}{3}, \frac{1}{3}$	$\frac{5}{3}, \frac{2}{3}$	
BB	1, 0	$\frac{4}{3}, 1$	4.
	3.		

1. AB is strictly dominated by BA
2. AA is also strictly dominated by BA
3. X is strictly dominated by Y
4. BB is strictly dominated by BA

Therefore,

$$NE = \{(BA, Y)\}$$

Problem 2

Find all *pure-strategy* Bayesian Nash equilibria of the following game:

- Nature determines whether the payoffs are as in Game 1 (with probability ρ) or as in Game 2 (with probability $1 - \rho$).
- Player 2 learns whether Nature has drawn Game A or Game B, but player 1 does not.
- Player 1 chooses either F or T ; player 2 simultaneously chooses either F or T .
- Payoffs are given by the game drawn by Nature.

Game A

1\2	F	T
F	3,1	0,0
T	0,0	1,3

Game B

1\2	F	T
F	3,0	0,1
T	0,3	1,0

Remember that only player 2 knows which game we are in, very important

And $\rho \in (0,1)$, so ρ cannot be 1 or 0

ρ and p are the same, I was just a little bit lazy when I answered the problem

The Bayesian matrix is

	FF	FT	TF	TT
F	$3, p$	$3p, 1$	$3 - 3p, 0$	$0, 1 - p$
T	$0, 3 - 3p$	$1 - p, 0$	$p, 3$	$1, 3p$

For example, in (F, FF) :

$$v_1 = p \cdot 3 + (1 - p) \cdot 3 = 3$$

$$v_2 = p \cdot 1 + (1 - p) \cdot 0 = p$$

And in (T, FF)

$$v_1 = p \cdot 0 + (1 - p) \cdot 0$$

$$v_2 = p \cdot 0 + (1 - p) \cdot 3 = 3 - 3p$$

Underlining BRs:

1 \ 2	FF	FT	TF	TT
F	<u>3</u> , <u>p</u>	3p, <u>1</u>	3-3p, 0	0, 1-p
T	0, 3-3p	1-p, 0	<u>p</u> , <u>3</u>	<u>1</u> , <u>3p</u>

So, for example, (T, TT) can never be an equilibrium since $3p < 3$ as $p \in (0, 1)$

This yields the following equation for (F, FT) to be a BNE

$$3p \geq 1 - p \Leftrightarrow \frac{1}{4} \leq p < 1$$

So, when $\frac{1}{4} \leq p < 1$, the BNE is (F, FT)

And for (T, TF) to be a BNE:

$$p \geq 3 - 3p \Leftrightarrow p \geq \frac{3}{4}$$

So, when $\frac{3}{4} \leq p < 1$, the BNEs are (T, TF) and (F, FT)

Why is, e.g., (F, FF) never a BNE? Well, in that case, T is a better response of player 2 since $p < 1$

And when $p < \frac{1}{4}$, no BNE exists

Problem 3

Don Vito wishes to sell a piece from his art collection. There are two potential buyers whose private values are independent and uniformly distributed over the set $\{0, 1, 2\}$. In other words, each buyer has three possible values. Don Vito will offer the piece using a second-price sealed-bid auction. As a quirk, he requires that all bids in the auction must be nonnegative integers.

- (a) What is Don Vito's expected revenue?
- (b) Now an adviser has suggested that Don Vito should set a *reserve price* $r > 0$. A positive reserve price modifies the auction as follows: If both bids are below r , then neither bidder obtains the art piece and it is destroyed. If both bids are at or above r then the usual second-price sealed-bid auction rules apply. If only one bid is at or above r then that bidder obtains the art piece and pays r .
 - b1) Is it still a weakly dominant strategy for each player to bid his valuation when $r > 0$? Explain briefly.
 - b2) What is the Don Vito's expected revenue when $r = 1$ (as suggested by the adviser)?

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- b3) What explains the difference between your answers in (a) and b2)?

a

In a sealed-bid action, players submit written bids without knowing the bid of the other players. The highest bidder wins but the price paid is the second-highest bid. In this type of auction, it is always a weakly-dominant strategy to bid one's own valuation (c.f. slides or

Proposition 1 *In a second price auction, it is a weakly dominant strategy to bid one's value, $b_i(s_i) = s_i$.*

Proof. Suppose i 's value is s_i , and she considers bidding $b_i > s_i$. Let $\hat{b}$ denote the highest bid of the other bidders $j \neq i$ (from i 's perspective this is a random variable). There are three possible outcomes from i 's perspective: (i) $\hat{b} > b_i, s_i$; (ii) $b_i > \hat{b} > s_i$; or (iii) $b_i, s_i > \hat{b}$. In the event of the first or third outcome, i would have done equally well to bid s_i rather than $b_i > s_i$. In (i) she won't win regardless, and in (ii) she will win, and will pay $\hat{b}$ regardless. However, in case (ii), i will win and pay more than her value if she bids $\hat{b}$, something that won't happen if she bids s_i . Thus, i does better to bid s_i than $b_i > s_i$. A similar argument shows that i also does better to bid s_i than to bid $b_i < s_i$. Q.E.D.

General formula is

$$B \cdot \frac{1}{9}$$

Where B stands for second-highest bid and $\frac{1}{9}$ since the probability of a player choosing each bid is uniformly distributed, i.e., $\frac{1}{3}$ for each outcome so $\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$ for, e.g., player 1 choosing 2 and player 2 choosing 1 since player 1 then pays second-highest bid, which is 1

1\2	0	1	2
0	0	0	0
1	0	$\frac{1}{9}$	$\frac{1}{9}$
2	0	$\frac{1}{9}$	$\frac{2}{9}$

Which implies that Don Vito's expected revenue is $R = \frac{5}{9}$

B1

The strategy is—as in a—still to offer one's own private value c.f. the same Proposition

B2

If, for example, player 1 bids 1 and player 2 bids 0, player 1 has to pay 1 rather than 0 because of the reserve price $r = 1$

$1 \backslash 2$	0	1	2
0	0	$\frac{1}{9}$	$\frac{1}{9}$
1	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
2	$\frac{1}{9}$	$\frac{1}{9}$	$\frac{2}{9}$

So, Don Vito's expected revenue when $r = 1$ is $R = \frac{9}{9} = 1$

B3)

The reserve price ensures that the price paid will never be less than 1. This reserve price does not change the player's strategy, so it has increased the expected revenue without changing the player's strategy.

Problem 4

There are two people, who want to produce a common public good through voluntary contributions. Simultaneously, every player i , $i = 1, 2$, contributes x_i . The amount of public

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good produced is

$$y = x_1 + x_2.$$

The payoff of each player i is

$$u_i = \theta_i y - y^2 - x_i.$$

where θ_i is a parameter privately known by player i , and θ_1 and θ_2 are independently and identically distributed with uniform distribution on $[1, 2]$. Assume that x_i can be positive or negative. Compute a symmetric Bayesian Nash equilibrium.

The FOC is

$$\begin{aligned}\frac{\partial}{\partial x_i}(u_i) &= \frac{\partial}{\partial x_i}(\theta_i y - y^2 - x_i) \\ &= \frac{\partial}{\partial x_i}(\theta_i(x_i + x_j) - (x_i + x_j)^2 - x_i) \\ &= \theta_i - 2(x_i + x_j) - 1 \\ &= 0\end{aligned}$$

Best response is then

$$\theta_i - 2(x_i + x_j) - 1 = 0 \Leftrightarrow x_i = \frac{\theta_i - 2x_j - 1}{2}$$

Player i expects $\mathbb{E}[\theta] = \frac{1+2}{2} = \frac{3}{2}$ since θ is uniformly distributed on $[1, 2]$

Therefore, average contribution is

$$x_i = \frac{\theta_i - 2x_j - 1}{2} \Rightarrow \bar{x} = \frac{\frac{3}{2} - 2\bar{x} - 1}{2} \Leftrightarrow \bar{x} = \frac{1}{8}$$

And each player in optimum then contributes:

$$x_i^* = \frac{\theta_i - 2x_i - 1}{2} = \frac{\theta_i - 2 \cdot \frac{1}{8} - 1}{2} = \frac{1}{2}\theta_i - \frac{5}{8}$$